

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: COMPUTER VISION COURSE CODE: ITA0515 Slot B

COURSE OUTCOME COVERED

CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5) Assessment Tool Weightage: 40%

QUESTIONS:5 Total Marks 50 Duration :60 Minutes Annexure A
Industry Problem Solving Task

Code of Conduct:

I PAVAN JANA (192472313) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *PAVAN J*

Rubrics for Evaluation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks(100)
Understanding of Industry Problem	20%	Comprehensive understanding of real world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context	
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts	
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution	
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools	

Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis	
Professionalism	5%	Highly professional, well documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation	

Analyze the given problem by comparing available techniques against the specified constraints and requirements. Evaluate and justify your decision using logical reasoning, clearly indicating how the chosen approach satisfies the conditions.

1. Traffic Surveillance Model Selection

A traffic monitoring system must:

- Accuracy $\geq 90\%$
- Latency ≤ 80 ms
- Works under varying lighting

Options:

Method A: 88%, 60 ms

Method B: 91%, 85 ms

Method C: 92%, 75 ms

Compare and evaluate which method satisfies all constraints. Justify logically.

2. Drone Vision System Constraint

A drone system requires:

- Lightweight computation
- Real-time detection (<60 ms)
- Accuracy $\geq 85\%$

Options:

Model A: 83%, 40 ms

Model B: 87%, 65 ms

Model C: 86%, 55 ms

Compare and evaluate the suitable model. Justify using constraints.

3. Security System Recognition

A security system must:

- High accuracy (>92%)
- Moderate computation

Options:

Traditional: 85%, low computation

Deep Learning: 94%, high computation

Hybrid: 91%, moderate computation

Compare and evaluate the best approach. Justify logically.

4. Motion Tracking in Sports Analytics

Requirements:

- Accurate motion tracking
- Real-time performance

Options:

Optical Flow: accurate, slow

Motion Estimation: fast, less accurate

Compare and evaluate the suitable approach. Justify using constraints.

5. Face Recognition System Design

Constraints:

- Accuracy $\geq 90\%$
- Works in low light
- Moderate computation

Options:

Traditional: 82%

Deep Learning: 93%

Hybrid: 89%

Compare and evaluate the best choice. Justify logically.

1. Traffic Surveillance Model Selection

Requirements

- Accuracy $\geq 90\%$
- Latency ≤ 80 ms
- Must work under varying lighting

Given

Method Accuracy Latency Evaluation

A	88%	60 ms	✗ Accuracy too low
B	91%	85 ms	✗ Latency too high
C	92%	75 ms	☑ Satisfies both

Method A

Method A has **88% accuracy**, which is below the required 90%.

Although its latency is excellent at **60 ms**, it fails the accuracy requirement.

Therefore, Method A is rejected.

Method B

Method B provides **91% accuracy**, satisfying the accuracy requirement.

However, its latency is **85 ms**, while the maximum allowed is 80 ms.

Therefore, Method B is rejected.

Method C

Method C provides:

- Accuracy = **92% $\geq 90\%$**
- Latency = **75 ms ≤ 80 ms**

Therefore, it satisfies both numerical constraints.

If Method C is also designed/trained to handle varying lighting conditions, it is the

most suitable choice.

Final Answer

Method C is the best choice. It is the **only method that satisfies both the accuracy and latency constraints.**

Justification: In a traffic surveillance system, both accuracy and response time are important. Method C provides high accuracy while remaining within the required latency limit.

2. Drone Vision System Constraint

Requirements

- Lightweight computation
- Real-time detection < 60 ms
- Accuracy $\geq 85\%$

Given

Model Accuracy Latency Evaluation

A	83%	40 ms	✗ Accuracy too low
B	87%	65 ms	✗ Too slow
C	86%	55 ms	☑ Satisfies constraints

Model A

Model A has very good latency:

40 ms $<$ 60 ms

However, accuracy is only **83%**, which is below the required 85%.

Therefore, it cannot be selected.

Model B

Model B has **87% accuracy**, which satisfies the accuracy requirement.

But latency is:

65 ms > 60 ms

Therefore, it does not meet the real-time requirement.

Model C

Model C has:

- Accuracy = **86% $\geq$ 85%**
- Latency = **55 ms < 60 ms**

It satisfies both numerical requirements.

Since the drone also requires lightweight computation, Model C should be preferred if its computational requirements are lower or acceptable compared with Model B.

Final Answer

Model C is the most suitable choice.

Justification: It provides sufficient accuracy while maintaining real-time performance below 60 ms. Model A is too inaccurate, while Model B is too slow.

3. Security System Recognition

Requirements

- Accuracy > **92%**
- Moderate computation

Given

Approach	Accuracy Computation Evaluation		
Traditional	85%	Low	✗ Accuracy too low
Deep Learning	94%	High	⚠ Accuracy good, computation high

Approach Accuracy Computation Evaluation

Hybrid 91% Moderate ✗ Accuracy below requirement

Traditional Method

Accuracy = **85%**

Required accuracy is **greater than 92%**.

Therefore:

85% < 92%

It fails the accuracy requirement.

Deep Learning

Accuracy = **94%**

This satisfies the accuracy requirement.

However, computation is **high**, while the system requires moderate computation.

So it does not fully satisfy the stated constraints.

Hybrid

Accuracy = **91%**

Although computation is moderate, accuracy is below the required level.

Therefore:

91% < 92%

It fails the accuracy requirement.

Final Answer

None of the three approaches completely satisfies both constraints.

However, **Deep Learning is the best approach if high accuracy is the primary requirement**, because it achieves 94%.

If moderate computation is absolutely mandatory, the system needs an **optimized deep-learning model**, such as:

- Model compression
- Quantization
- Pruning
- Knowledge distillation
- Lightweight neural networks

Important Exam Point

Do **not** simply answer Hybrid because it has moderate computation. It fails the accuracy requirement.

Likewise, Deep Learning satisfies accuracy but violates the computation constraint.

Therefore, the logically correct answer is: no existing option fully satisfies all constraints; an optimized Deep Learning approach would be required.

4. Motion Tracking in Sports Analytics

Requirements

- Accurate motion tracking
- Real-time performance

Given

Method	Accuracy	Speed	Evaluation
Optical Flow	High	Slow	⚠ Accurate but not real-time
Motion Estimation	Lower	Fast	⚠ Real-time but less accurate

Optical Flow

Optical Flow estimates the apparent movement of pixels between consecutive frames.

Advantages:

- High accuracy

- Detailed motion information
- Good for tracking complex movement

Disadvantage:

- Computationally expensive
- Can be too slow for real-time sports analytics

Motion Estimation

Motion Estimation is computationally faster.

Advantages:

- Fast
- Suitable for real-time processing
- Lower computational requirements

Disadvantage:

- Less accurate than Optical Flow
- May miss fine movements

Evaluation

The requirements contain **two competing objectives**:

Accuracy + Real-time performance

Neither method perfectly satisfies both.

If real-time performance is a **strict requirement**, Motion Estimation is preferable because it is faster.

If accurate motion tracking is more important and some latency is acceptable, Optical Flow is better.

Final Answer

Motion Estimation is generally the more suitable choice for real-time sports analytics, because sports applications require rapid processing.

However, a **hybrid approach** can provide better results: use fast Motion Estimation for normal tracking and Optical Flow selectively when highly accurate motion information is required.

5. Face Recognition System Design

Requirements

- Accuracy $\geq 90\%$
- Works in low-light conditions
- Moderate computation

Given

Approach	Accuracy	Low-Light Capability	Computation
Traditional	82%	Poor	Low
Deep Learning	93%	Good*	High
Hybrid	89%	Better	Moderate

*Assuming the deep-learning model is trained with suitable low-light/augmented data.

Traditional Method

Accuracy = **82%**

Required accuracy = **90%**

Therefore:

82% < 90%

It fails the accuracy requirement.

Traditional methods also tend to struggle with significant changes in lighting.

Deep Learning

Accuracy = **93%**

Therefore:

$$93\% \geq 90\%$$

It satisfies the accuracy requirement.

Deep-learning models can also learn robust facial features and can be trained using:

- Low-light images
- Brightness augmentation
- Contrast enhancement
- Different illumination conditions

The main disadvantage is **high computational cost**, which conflicts with the moderate-computation requirement.

Hybrid Method

$$\text{Accuracy} = 89\%$$

Required:

$$\geq 90\%$$

Therefore:

$$89\% < 90\%$$

It fails the accuracy requirement despite having moderate computation.

Final Answer

Deep Learning is the best choice based on recognition accuracy and low-light robustness, but its high computational cost means it does not completely satisfy the moderate-computation constraint.

A better practical solution would be a **lightweight deep-learning model optimized for low-light face recognition**.

Possible techniques include:

- Model pruning

- Quantization
- Knowledge distillation
- Lightweight CNN architectures
- Low-light image enhancement